Integrated Municipal Civil Registration and Digital Archive Management System

Abstract

The Integrated Municipal Civil Registration and Digital Archive Management System is a web-based e-Governance application developed to digitize and streamline the complete lifecycle of birth and death certificate management within municipal administrations. The system enables citizens to apply for birth and death certificates online, upload supporting documents, track application status, and download digitally generated certificates without repeatedly visiting municipal offices.

The application provides dedicated portals for Citizens, Municipal Officers, and Administrators, ensuring secure role-based access and efficient workflow management. Municipal officers can verify applications, approve or reject requests, and generate digitally signed PDF certificates with unique QR codes for instant authenticity verification. To improve data quality, the system includes duplicate record detection, preventing multiple registrations of the same event. A digital archive management module securely stores certificates and records, allowing quick retrieval through advanced search filters based on certificate number, applicant details, registration date, hospital, and other relevant information.

To enhance administrative efficiency, the system incorporates Service Level Agreement (SLA) monitoring, which tracks application processing time, identifies delayed requests, and automatically escalates overdue applications to higher authorities. Additionally, a comprehensive Audit Trail records every system activity, including application submission, verification, approval, modification, certificate generation, and download, thereby improving transparency, accountability, and compliance.

The system is developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js), with JWT for secure authentication, bcrypt for password encryption, Multer for document uploads, PDFKit/pdf-lib for certificate generation, QR Code integration for certificate verification, and Nodemailer for email notifications.

By integrating secure certificate issuance, intelligent record management, digital archiving, administrative monitoring, and accountability features into a single platform, the proposed system aims to improve service delivery, reduce manual effort, minimize processing delays, enhance record accuracy, and provide citizens with a faster, more transparent, and reliable municipal certificate management experience.